hru-data.hru

Purpose

`hru-data.hru` is the pointer table for every HRU in the project. Each record names one HRU and lists the parameter sets it draws from: topography, hydrology, soils, land use management, soil and plant initialization, surface storage, snow, and field. The names in this file are looked up in the corresponding parameter files (`topography.hyd`, `hydrology.hyd`, `soils.sol`, `landuse.lum`, `plant.ini`, `snow.sno`, `field.fld`). The integer indices resolved by those lookups are stored in `hru_db(i)%dbs` and used throughout the simulation to fetch HRU properties.

If the file is missing or set to `null` in `file.cio`, the reader allocates `hru_db(0:0)` and no HRU data is loaded.

Source

- Reader: src/hru_read.f90
- Type definition: src/hru_module.f90 (`type hru_databases_char`, `type hydrologic_response_unit_db`)
- File name pointer: src/input_file_module.f90 (`in_hru%hru_data`, default `"hru-data.hru"`)

Format

- Line 1: title. Skipped.
- Line 2: header. Skipped.
- Line 3 and beyond: one record per HRU. The reader makes two passes. The first pass scans the file to find the largest `id` and allocates `hru_db(0:imax)`. The second pass reads each record list-directed into `k, hru_db(i)%dbsc`, where `k` is a throwaway read of the id and `i` is the id parsed from the same line.

Record layout: 10 list-directed values per HRU.

#	Column	Maps to	Type	Description
1	id	record index `i`	int	HRU number. Used as the array index into `hru_db`. Does not have to be sequential. The largest `id` in the file sets the array size.
2	name	`hru_db(i)%dbsc%name`	char(40)	HRU name. Free-form identifier.
3	topo	`hru_db(i)%dbsc%topo`	char(40)	Topography record name. Looked up in `topography.hyd`. Missing names are logged to `diagnostics.out` as "not found (topography.hyd)".
4	hydro	`hru_db(i)%dbsc%hyd`	char(40)	Hydrology record name. Looked up in `hydrology.hyd`. Missing names are logged as "not found (hydrograph.hyd)".
5	soil	`hru_db(i)%dbsc%soil`	char(40)	Soil record name. Looked up in `soils.sol`. Missing names are logged as "not found (soils.sol)".
6	lu_mgt	`hru_db(i)%dbsc%land_use_mgt`	char(40)	Land use and management record name. Looked up in `landuse.lum`. Missing names are logged as "not found (landuse.lum)".

#	Column	Maps to	Type	Description
7	soil_plant_init	`hru_db(i)%dbsc%soil_plant_init`	char(40)	Soil and plant initialization record name. Looked up in `plant.ini`. Missing names are logged as "not found (plant.ini)". A match also cross-walks the initial soil-test, pesticide, pathogen, heavy-metal, salt, and constituent records named inside the `plant.ini` entry.
8	surf_stor	`hru_db(i)%dbsc%surf_stor`	char(40)	Surface storage (reservoir or wetland) record name. The reader stores the name but does not resolve a pointer for it in `hru_read`. Use `null` if the HRU has no surface storage.
9	snow	`hru_db(i)%dbsc%snow`	char(40)	Snow record name. Looked up in `snow.sno`. The literal `null` is allowed and suppresses the not-found warning.
10	field	`hru_db(i)%dbsc%field`	char(40)	Field record name. Looked up in `field.fld`. The literal `null` is allowed and suppresses the not-found warning.

The reader does not enforce a count line. The number of HRUs comes from the maximum `id` seen during the scan pass, so a gap in the numbering allocates unused slots in `hru_db`.

Example

`refdata/Ames_sub1/hru-data.hru`:

```
hru-data.hru:
      id  name                      topo            hydro           soil           lu_mgt   soil_
       1  hru0001              topohru0001          hyd0001       soil_01-h1      cosy_lum        
       2  hru0002              topohru0002          hyd0002       soil_02         mntill_corn_lum 
       3  hru0003              topohru0003          hyd0003       soil_03-h3      cosy_lum        
       4  hru0004              topohru0004          hyd0004       soil_04         cosy_lum        
```

Notes on the example:

- HRU 2 uses `mntill_corn_lum` while the rest use `cosy_lum`. The land use record drives tillage, plant cover, and management operations.
- `surf_stor` and `field` are `null` for every HRU. None of the HRUs has a pond, wetland, or tile-drained field assigned.
- `snow001` is the same record for every HRU. The snow record is a per-elevation-band parameter set.

Related files

- `object.cnt`. The `hru` count must be at least the largest `id` in this file.
- `hru.con`. One row per HRU, in the same id order.
- `topography.hyd`, `hydrology.hyd`, `soils.sol`, `landuse.lum`, `plant.ini`, `snow.sno`, `field.fld`. Targets of the name pointers above.
- `file.cio`. The `hru` category line points to this file.